

An Automated CNN-Based Model for Leukaemia Diagnosis from Blood Cell Images

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Abstract— The microscopic blood smear images have proven difficult in the accurate diagnosis of leukemia because of the finer morphological changes and also because of the requirement of manual examination. Deep learning methods availed by automation have the potential of enhancing the reliability and efficiency of diagnosis. The convolutional neural network structure is developed to classify the multiclass blood cell by using hierarchical feature extraction. The model combines convolutional layers, batch normalization, max pooling, and dropout to promote the stability in learning and curb overfitting. Training uses real-time data augmentation and normalization and adaptive learning rate scheduling is used to enhance convergence. The experimental data is collected as 3,242 labelled blood cell images that are arranged as benign, malignant pre-B, malignant pro-B, and malignant early pre-B. Experimental analysis shows the good classification performance, where the overall accuracy is 97.9. The values of precision, recall and F1-score are above 0.94 in all classes and very little misclassification has occurred. The trends of training and validation show that convergence remains constant with minimal loss in generalization. Such results promote clinical translation.

Keywords— *Leukemia detection, blood cell classification, convolutional neural network, deep learning, medical image analysis, automated diagnosis*

I. INTRODUCTION

Leukemia is a terminal haematological cancer due to uncontrolled increase in white blood cells, and the early and precise diagnosis of the disease is important in enhancing the survival chances and treatment protocols. The use of microscopic analysis of peripheral blood and bone marrow samples is also a conventional method of diagnosis, but this procedure is time consuming, laborious and extensively relies on the ability of skilled interpreters, which predisposes this subject to subjective errors. In an attempt to overcome these shortcomings, computer-aided diagnostic systems which are founded on analysis of medical imaging have become a topic of great concern over recent years. Large-scale comparison and benchmarking of automated leukemia detection and classification algorithms have been made possible through the availability of standardized and well-annotated datasets like the Acute Lymphoblastic Leukemia Image Database [1].

Conventional machine learning methods have also been extensively studied to achieve this objective, as multiple studies have incorporated better algorithms to achieve a higher level of diagnostic accuracy besides lowering the complexity of computation [2]. The unsupervised clustering algorithms, especially the k-means, have proven to be effective in clustering blood cell images which are characterized in terms of morphology and thus assist in the initial phases of automatic leukemia detection [3]. It has been further improved by clustering with supervised classifiers like the support vectors machines which has made it possible to discriminate malignant and benign cells more accurately in the microscopic images [4]. As deep learning progressed, recent studies have moved in the direction of convolutional neural network-based models with the ability to learn hierarchical features directly upon raw image data and achieve better robustness and classification results across a wide range of data [5].

Irrespective of these improvements, issues to do with the blood cell image segmentation have remained because of the varying staining, light illumination, overlapping cells and background noise, which directly impact the features extraction and classification reliability [6]. The challenges consist of the need to deal with them with powerful preprocessing pipelines and adaptable learning models that can be applied across clinical situations. Also, the introduction of intelligent medical Internet of Things technologies has come up with something new, such as automated diagnostics of microscopic images in real-time, the analysis in the cloud and remote clinical decision support, which has increased the access and efficiency of leukemia diagnostics [7]. Taken together, these advancements underscore the increasing significance of automated and precise, scalable diagnostic methods to leukemia, and the necessity of more investigations aimed at integrating emerging and innovative image processing, machine learning, and smart healthcare technologies to enhance patient outcomes in terms of the ability to promptly detect the condition, lessen the diagnostic workload, and enhance overall care results [1]-[7].

II. LITERATURE REVIEW

The image analysis literature of automated blood cancer detection shows a consistent trend of transitioning to the state of sophisticated deep learning architectures. Saritha et al. have provided an in-depth review of the microscopic analysis of blood samples, they discussed publicly available and laboratory prepared datasets that are used in the detection of leukemia and the various methodologies that have been employed in the detection and analysis of the data which all involve classical image preprocessing, feature extraction and segmentation, the author concluded that automated systems have a significant effect in reducing diagnostic variation, but at the cost of not providing any single unified accuracy value [8]. Kumar et al. have applied white blood cancer detection with bone marrow microscopic images and the use of convolutional neural network architecture that is trained on augmented images and experimental results showed that classification accuracy was over 97 percent that emphasizes the importance of deep learning compared with other feature-based methods [9]. Vinayagam et al. suggested an adapted hybrid learning scheme based on hospital-acquired blood cell images as a combination of a medical image processing method with an IoT-based alert system in which handcrafted features and learned representations were used together to produce an accuracy of 98.2 per cent and proved to be suitable in real-time telemedicine applications [10].

Nasra proposed a convolutional neural network based pneumonia detection on the datasets of chest X-ray images, with preprocessing based on contrast enhancement and noise reduction, and the model showed high diagnostic accuracy, demonstrating the versatility of CNN-based models to various medical imaging tasks outside the field of hematology [11]. The authors Saeed et al. trained a deep learning-based diagnostic model that identified acute lymphoblastic leukemia using large and well-labelled blood smear images datasets, utilizing the high-capacity architectures, like EfficientNet, and achieved 99.73% accuracy, which is the state-of-the-art performance of the deep-learning-based leukemia classifier [12]. Selvakumaran et al. developed a hybrid deep learning system based on deep laboratory-generated blood smears, which applied convolutional neural networks and ensemble learning approaches, and was optimized via grid search achieved an accuracy of 98.9 tying in greater robustness than single models [13]. All these studies lead to an inference that the quality of datasets, depth of architecture, optimization schemes, and consistency of preprocessing have direct positive effects on the diagnostic reliability, scalability as well as scalability on real life clinical medical imaging setting. Besides, Nasra introduced an EfficientNetV2-L model with a fine-tuning technique to detect melanoma based on datasets of dermoscopic images, in which transfer learning and fine-tuning of the model obtained high classification rates, which validates the efficacy of deep learning and model fine-tuning in the context of complicated medical image classification problems and justifies their use in the blood cancer diagnostics field [14].

III. METHODOLOGY

In this research, a developed convolutional neural network is used to categorize the images of blood cells related to leukemia. It will be the nature of the proposed architecture to capture and

analyse delicate morphological patterns that exist in the microscopic blood cell images. The data utilized in this study is of four different classes such as the benign cells, malignant pre-B cells, malignant pro-B cells, and malignant early pre-B cells. To have balanced learning and credible performance evaluation, images labelled in each category are put in different directories and the entire dataset is thoroughly subdivided into training subset, validation subset and testing subset. To enhance the strong stability and the ability to generalize the model, data augmentation and normalization methods are used in real-time to enhance the training process. It uses the convolutional layers and batch normalization, pooling and dropout layers in network architecture and trains with 30 cycles by adaptive learning rate strategy. The accuracy metrics and confusion matrix analysis are used to assess model performance with accuracy and loss curves used to monitor training behaviour. The effectiveness of the model to differentiate the types of leukemia-related blood cells is discussed in the following section.

A. Dataset Description

Within this approach, the data is widely grouped in benign and malignant groups with the latter further subdivided into the malignant early Pre-B, malignant Pre-B, and malignant Pro-B subgroups as shown in Fig. 1. The entire data will include 3,242 images, which are divided into training, validation, and testing data sets of 56, 14, and 30 percent of the data, respectively. This distribution is effective as it will guarantee adequate sample points to train successfully the model, and will also maintain a high amount of unknown samples on which the model will be determined to be effective in evaluating its performance and assessing its ability to generalize to new and previously unknown samples.

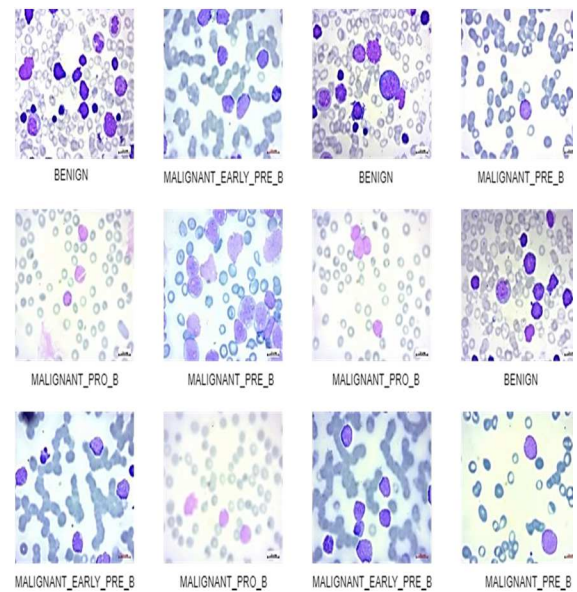


Fig. 1. Dataset Description

B. Dataset Pre-processing

Image classification tasks are broadly done by convolutional neural networks because they are efficient at addressing the high computational requirements of image data. In this work, all images provided are resized to 244x 244 pixels uniformly to provide the same processing throughout the network. In the first

stage of preprocessing, pixel intensity values are rescaled to range [?]1 to 1, a factor that leads to enhanced training stability and convergence rate. Moreover, data augmentation methods, including rotation, shifting, zooming, and flipping are used to increase the capacity of the model in generalizing to various image variations and reducing overfitting as shown in Fig. 2. The process of the training is also efficient as the randomly generated batches of the pictures are eight images, which means that learning is efficient, and the computation process is easy. The fact that the CNN model explicitly defines the input and output layers allows it to be trained and analysed systematically and properly classify the data as an image.

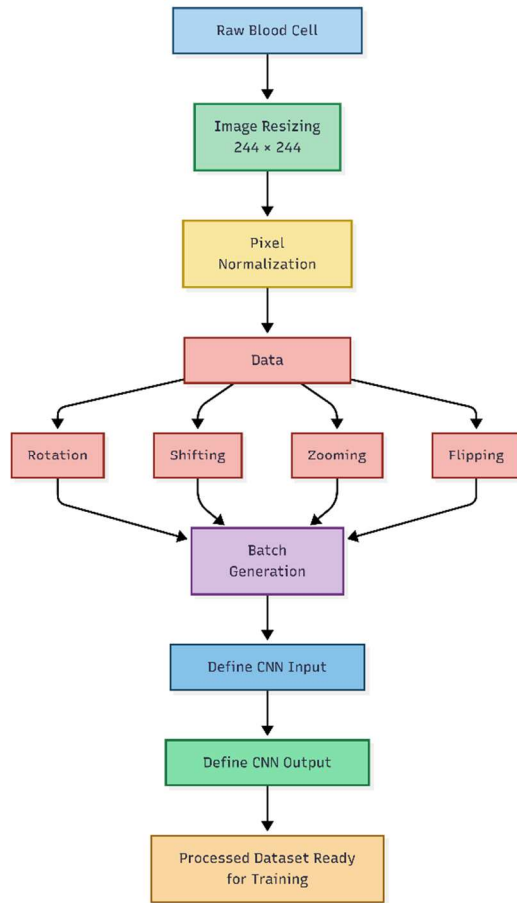


Fig. 2. Dataset Pre-processing Workflow

C. Proposed Methodology

The convolutional neural network proposed in Fig. 3 will be used to categorize the leukemia-related blood cell images in a

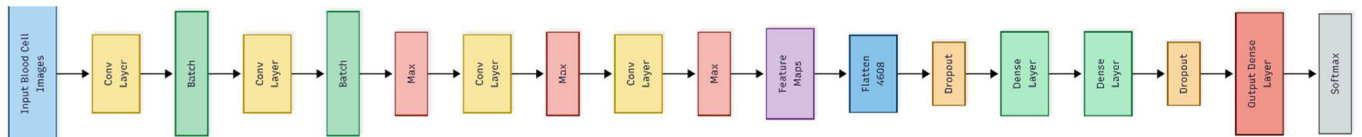


Fig. 3. Proposed CNN Model

IV. RESULTS

The findings of the present paper prove that the convolutional neural network designed can be successfully used

clear progress of layers that will enable easy figuring out of features and a precise classification. The architecture starts with a convolutional layer of 128 filters, and then the activation distributions are stabilised by using the batch normalization which helps to accelerate learning. This is followed by a second convolutional block of 256 filters and once again, it is combined with batch normalization to preserve activation invariance and reduce internal covariate shift. A max pooling layer is subsequently adopted in order to decrease the spatial dimensionality and maintain the most informative features. The next phases involve the interchange of 256 and 512-filter convolutional layers with the intervening maximum pooling operations at successive stages that gradually down sample feature maps, allowing the network to learn more abstract and high-level features of the original input images.

The feature maps generated at the convolutional pipeline end point are 3 x 3 x 512-dimensional, but they are flattened to a one-dimensional feature containing 4,608 units, which captures salient properties of the blood cell images. This is the part of the network where the input is fed by this vector. In order to minimize the chances of overfitting and enhance the performance of generalization, the dropout layer is used at once after the flattening layer. The network is then followed by a dense layer of 1,024 neurons that learns the interaction of complex features followed by a second dense layer of the same size with an additional dropout layer that also performs an extra and more effective regularization step by randomly deactivating a portion of the neurons during training.

A dense output layer of four neurons that represent the target categories of benign, malignant pre-B, malignant pro-B and malignant early pre-B cells completes the classification stage. To be able to make predictions that are interpretable and reliable, a softmax activation function is used to convert the output scores into the class-wise probability distributions. Overall, the model consists of 15,611,504 parameters, including 15,605,124 parameters that can be trained, a value of depth and representational capacity of the network. The architectural design uses the known deep learning principles of medical image analysis by combining convolutional, pooling, batch normalization, dense, and dropout blocks. The combination allows learning both local and global features effectively, training them in a stable, efficient manner, and being highly capable of generalization with high accuracy and recollection rates across all classes.

to classify blood cell images related to leukemia. As can be seen in both training and evaluation datasets, the high performance observed was steadily high, which means that the model has

acquired the discriminatory features that help it to correctly classify. The model had a recall of 93.4, a precision of 91.6 and a general classification accuracy of 97.9 on the test dataset. The training was done in thirty epochs with batches of images of 32 elements. The quantitative outcomes and detailed performance studies are shown in the following sections.

A. Accuracy and Loss Plots

The accurateness and loss curves used in the CNN-based blood cell image classification task after 30 training epochs are shown in Fig. 4 and 5. The accuracy plot with the range of 40 percent to almost 100 percent indicates that the accuracy of the training and validation increases steadily and gradually, which is a good learning behavior. According to this trend, the model can produce the relevant patterns in the data rather quickly and adapt effectively to unusual samples. In line with this, the loss curve start at a relatively higher value and then drops drastically towards zero as the training proceeds which means that there is a constant change towards an increased accuracy in the prediction. The strong correlation of the training and validation curves of the accuracy and loss values show that the learning is stable and that the model can generalize and not show a high degree of overfitting. All in all, these performance patterns prove the idea that the proposed CNN is an efficient learner of discriminative features and a correct classifier of blood cell images related to leukemia.

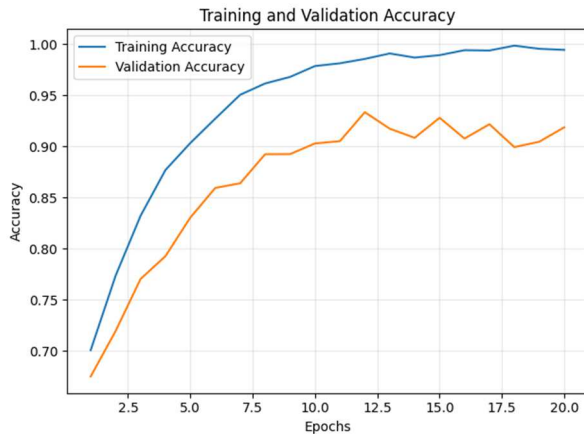


Fig. 4. Training and Validation Accuracy Plot

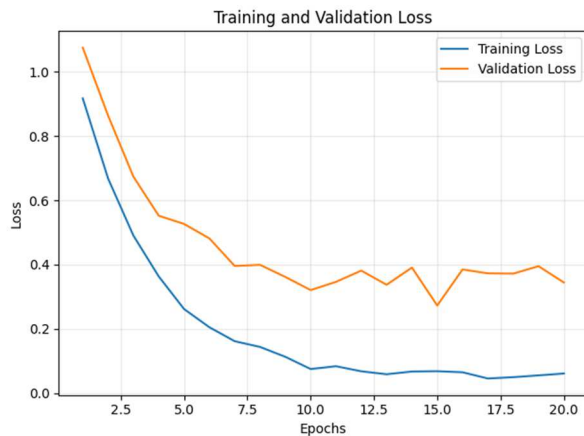


Fig. 5. Training and Validation Loss Plot

B. Confusion matrix

The confusion matrix in Fig. 6 is utilized to assess the performance of the proposed model in categorizing blood cell images related to leukemia in three and benign classes. The findings show that there is a high classification ability with few misclassifications. With benign samples, 171 cells were correctly classified, and there was one case that was classified as wrongly malignant pre-B and one case that was classified as wrongly malignant pro-B. The model proved to be very accurate in identifying malignant pre-B cells with only a handful of errors, as the malignant pro-B cells were correctly identified in all instances (223). In the malignant early pre-B group, 279 samples were rightly identified and there was little confusion where two samples classified as malignant pro-B and ten samples were benign. Altogether, the confusion chart demonstrates that the model is consistent and reliable, helping to confirm the fact that it is effective in the process of differentiating various blood cell types related to blood-cancer malignancies and leukemia, in particular.

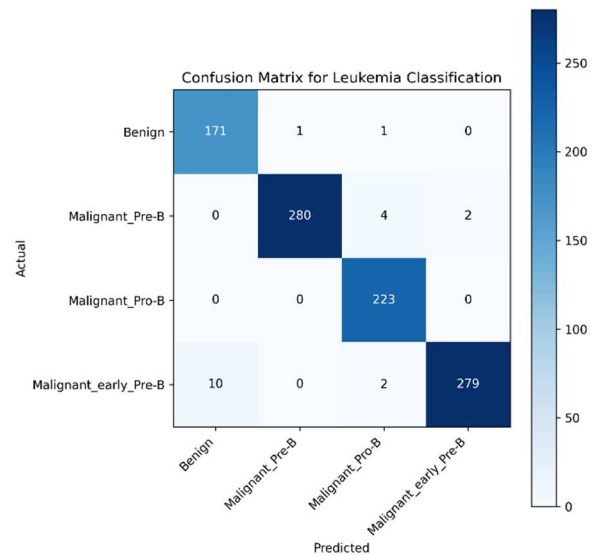


Fig. 6. Confusion Matrix

C. Classification Report

The comparison of all classes indicates that the given model has a high potential in recognizing the benign and malignant types of blood cells since the accuracy, recall, and the F1-score values in Table I are high. The perfect accuracy and the F1-score of the malignant pre-B class showed that there are no false positive predictions made of this group. Moreover, benign samples were also detected with high levels of reliability and the recall value was at 0.99. The high accuracy and specificity rates within all classes also bolsters the strength of the model with the only significant difference of a slight decrease in the recall of 0.96 in the malignant early pre-B category. All these performance measures show the stability and consistency of the model in identifying various blood cell subtypes. In general, the obtained accuracy of 97.9 percent highlights the usefulness of the model to facilitate accurate and credible leukemia diagnosis.

Table I: Classification Report

Name of the class	Precision	Recall	F1-Score
Benign	0.94	0.99	0.97
Malignant Pre-B	1.00	0.98	0.99
Malignant Pro-B	0.97	1.00	0.98
Malignant early Pre-B	0.99	0.96	0.98

V. CONCLUSION

The suggested network exhibits great results regarding the classification of blood cell images associated with leukemia with an overall accuracy of 97.9. Accuracy and loss curves show that there is a steady point to the model which is characterized by high accuracy and also a large decrease in the loss values. Although these results were positive, they point to the possibility of overfitting, which places the necessity of enhancing the generalization capabilities. Accurate classification is shown in the confusion matrix in all four categories with a few misclassifications especially in the malignant pre-B which had a perfect F1-score of 0.99. The classification report also supports the strength of the model since the values of precision, recall and F1-score are all above 0.94. However, further improvements to the model are needed to make it even more robust, such as the use of more sophisticated regularization methods, testing on a wider range of datasets, and testing other network designs. Besides, the deployment strategies should be enhanced to facilitate realistic applicability and to prolong the applicability of the model in the real diagnosis settings.

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